

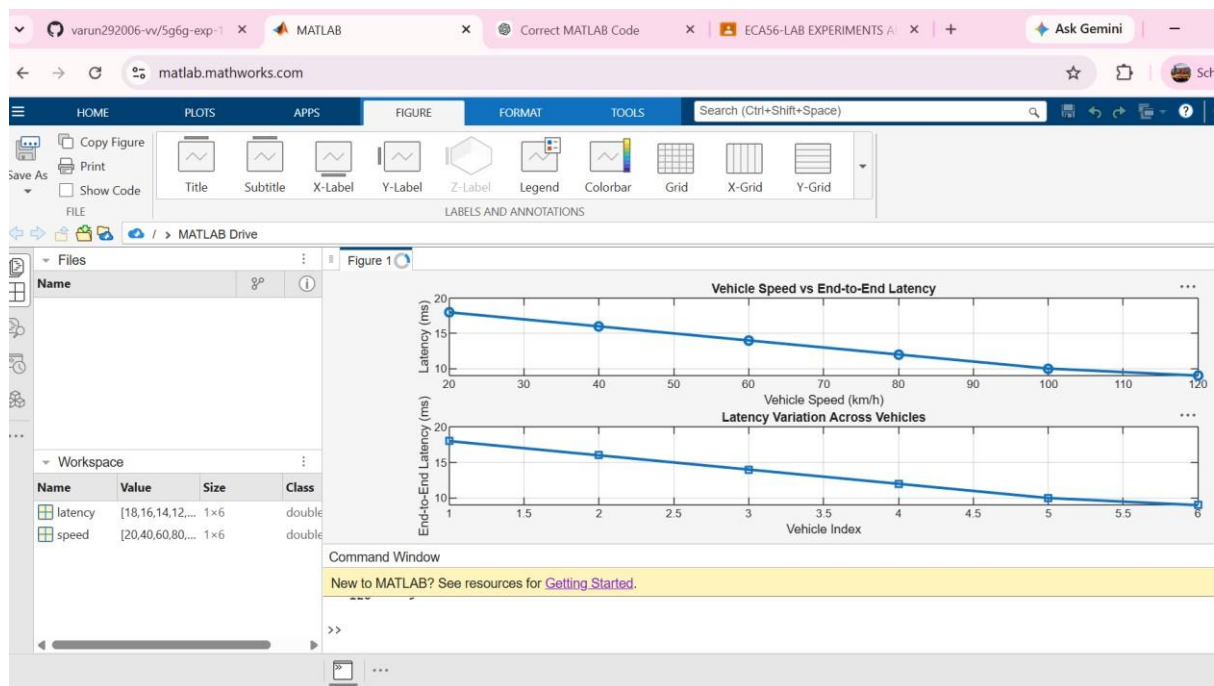
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EXPERIMENT 13

OBJ 1

```
clc; clear;
close all;
% Vehicle Speed (km/h) speed
= 20:20:120;
% Simulated End-to-End Latency (ms) latency = [18
16 14 12 10 9]; disp('Vehicle Speed (km/h) End-to-
End Latency (ms)') disp([speed' latency']) figure
% ----- Graph 1 ----- subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)') grid on
% ----- Graph 2 ----- subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index') ylabel('End-to-End
Latency (ms)')
grid on
```



OBJ 2

clc; clear;

close all;

% Vehicle Speed (km/h) speed

= 20:20:120;

% Simulated Packet Delivery Ratio (%) PDR = [99 98

97 95 93 90]; disp('Vehicle Speed (km/h) Packet

Delivery Ratio (%)') disp([speed' PDR']) figure

% ----- Graph 1 ----- subplot(2,1,1)

plot(speed,PDR,'-o','LineWidth',2)

title('Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)') grid

on

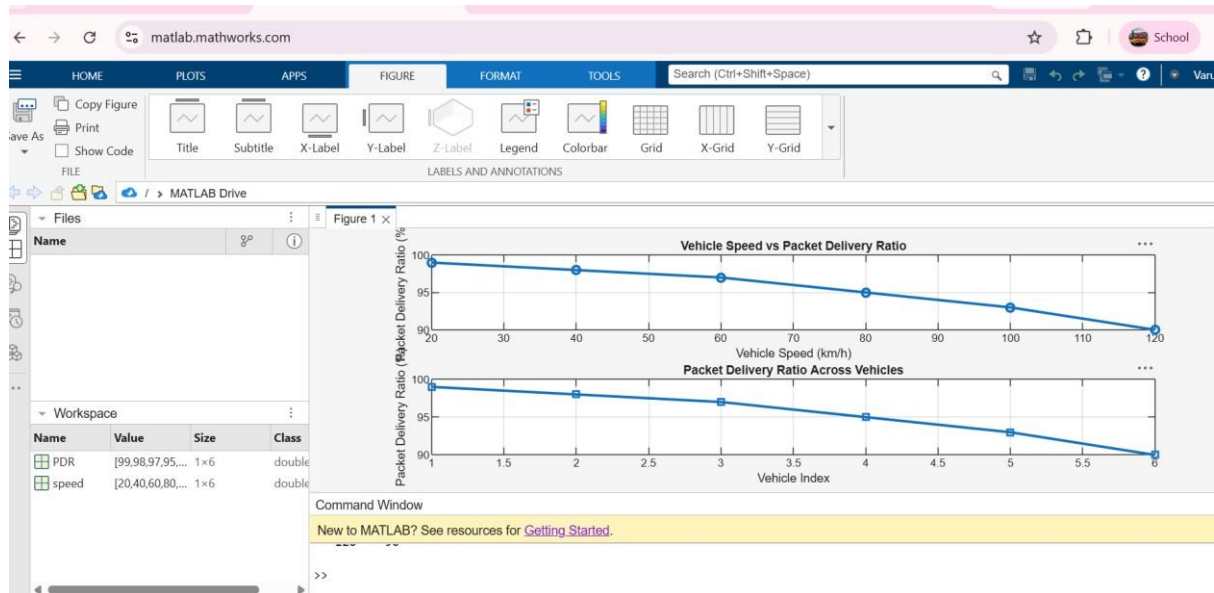
% ----- Graph 2 ----- subplot(2,1,2)

plot(1:length(PDR),PDR,'-s','LineWidth',2)

title('Packet Delivery Ratio Across Vehicles')

xlabel('Vehicle Index') ylabel('Packet

Delivery Ratio (%)') grid on



OBJ 3

clc; clear;

close all;

% Vehicle IDs vehicle

= 1:6;

% Vehicle Speed (km/h) speed

= [30 45 60 75 90 105]; %

Communication Range (m)

range = [280 260 240 220 200

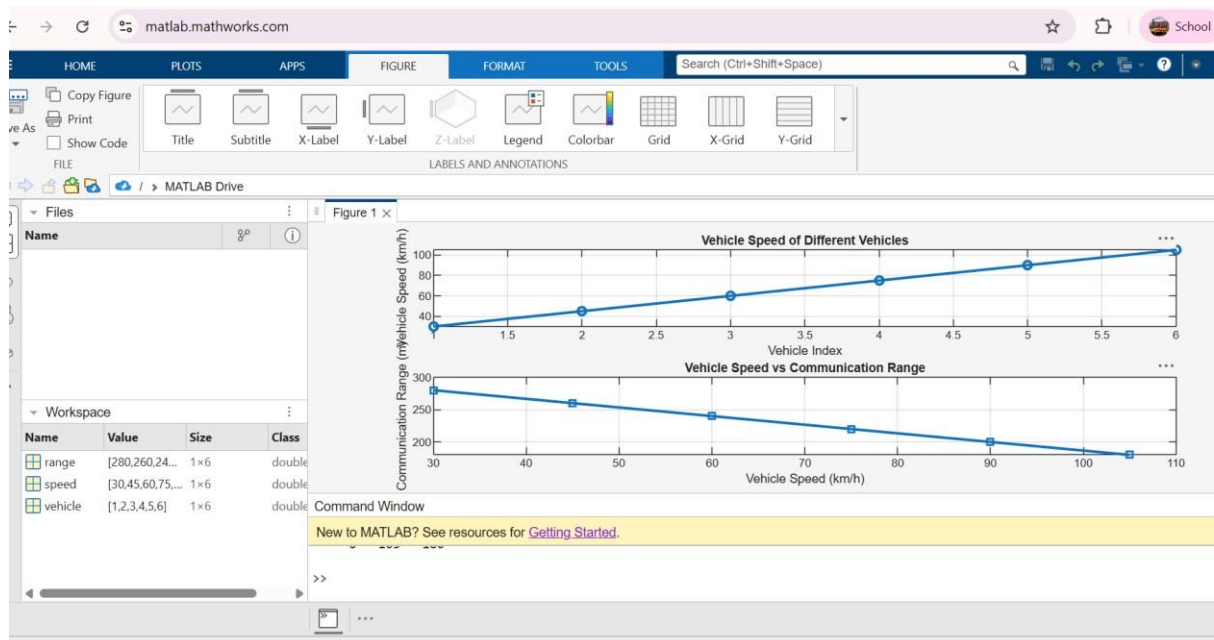
180]; disp('Vehicle

Speed(km/h) Communication

Range(m)')

disp([vehicle' speed' range']) figure

```
% ----- Graph 1 ----- subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index') ylabel('Vehicle
Speed (km/h)') grid on
% ----- Graph 2 ----- subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)') grid on
```



EXPERIMENT 14

OBJ 1

```
clc; clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
```

```

% Maximum Throughput (Mbps)
Max_TP = 100;
% Throughput calculation
Throughput = Max_TP*(1-exp(-SNR/10));
disp('SNR(dB) Throughput(Mbps)')
disp([SNR' Throughput']) figure
% ----- Graph 1 ----- subplot(2,1,1)
plot(SNR,Throughput,'-
o','LineWidth',2) title('5G Throughput
vs SNR') xlabel('Signal-to-Noise Ratio
(dB)') ylabel('Throughput (Mbps)') grid
on
% ----- Graph 2 -----

subplot(2,1,2) bar(Throughput)
title('Throughput at Different SNR Levels')
xlabel('SNR Index') ylabel('Throughput
(Mbps)') grid on

```



OBJ 2

```

clc; clear;

close all;

% SNR
values
(dB)

SNR = 0:5:30;

% Simulated Bit Error Rate (BER)
BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)
Reliability = (1 - BER) * 100;

% Display results disp('SNR(dB)   BER
Reliability(%'); disp([SNR' BER'
Reliability']);

figure;

% ----- Graph 1: SNR vs BER -----
subplot(2,1,1);

semilogy(SNR, BER, '-o', 'LineWidth', 2);
title('SNR vs Bit Error Rate in 5G');
xlabel('Signal-to-Noise Ratio (dB)');
ylabel('Bit Error Rate (BER)'); grid on;

```

% ----- Graph 2: SNR vs Reliability -----

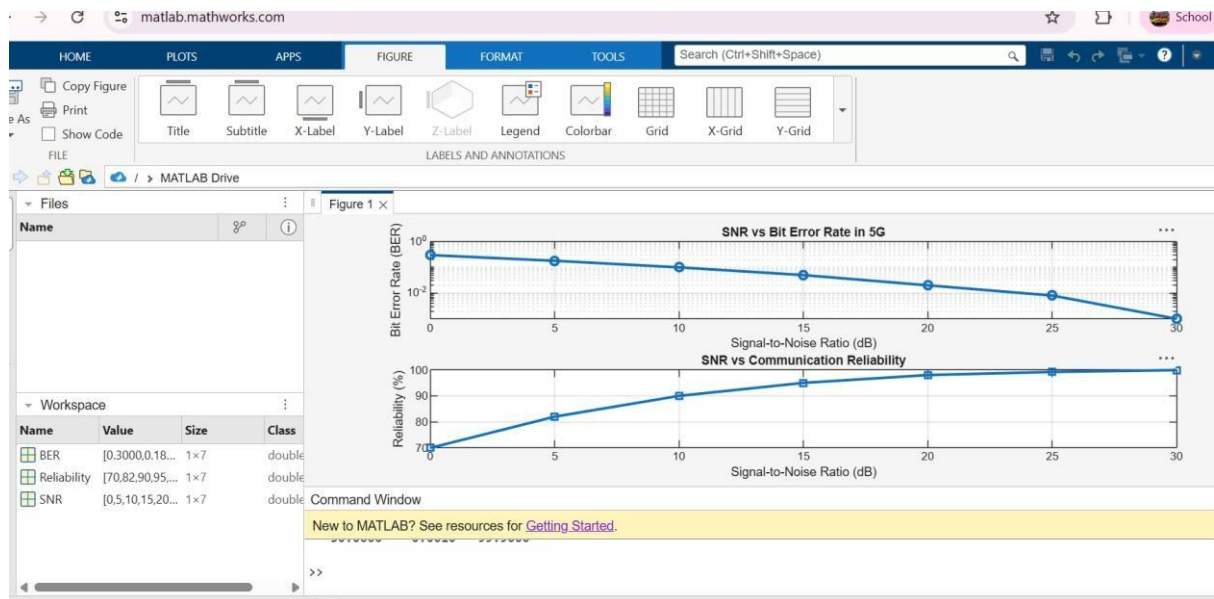
subplot(2,1,2);

plot(SNR, Reliability, '-s', 'LineWidth', 2);

title('SNR vs Communication Reliability');

xlabel('Signal-to-Noise Ratio (dB)');

ylabel('Reliability (%)'); grid on;



OBJ 3

clc; clear;

close all;

% Modulation schemes

Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};

% Required SNR (dB)

SNR = [5 10 18 25];

% Spectral Efficiency (bits/s/Hz) Efficiency = [2 4

6 8]; disp('Required SNR (dB) and Spectral

Efficiency')

disp(table(Modulation',SNR',Efficiency', ...

'VariableNames',{'Modulation','SNR_dB','Efficiency'})) figure

% ----- Graph 1 ----- subplot(2,1,2)

plot(SNR,Efficiency,'-o','LineWidth',2)

title('SNR vs Spectral Efficiency')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Spectral Efficiency (bits/s/Hz)')

grid on

% ----- Graph 2 ----- subplot(2,1,1) plot(1:4,SNR,'-

s','LineWidth',2) xticks(1:4) xticklabels(Modulation)

title('Required SNR for Different Modulation Schemes')

xlabel('Modulation Scheme') ylabel('Required SNR

(dB)') grid on

